# 1. Dataset Information

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

# 2. Missing Values Before Cleaning

| | Missing Values | Percentage |
|---|---|---|
| PassengerId | 0 | 0.000000 |
| Survived | 0 | 0.000000 |
| Pclass | 0 | 0.000000 |
| Name | 0 | 0.000000 |
| Sex | 0 | 0.000000 |
| Age | 177 | 19.865320 |
| SibSp | 0 | 0.000000 |
| Parch | 0 | 0.000000 |
| Ticket | 0 | 0.000000 |
| Fare | 0 | 0.000000 |
| Cabin | 687 | 77.104377 |
| Embarked | 2 | 0.224467 |

# 3. Missing Values After Cleaning

```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Embarked       0
dtype: int64
```

```
Total missing values: 0
Updated dataset shape: (891, 11)
```

# 4. Categorical Encoding

| | PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Embarked_C | Embarked_Q | Embarked_S |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 0 | 3 | Braund, Mr. Owen Harris | 0 | 22.0 | 1 | 0 | A/5 21171 | 7.2500 | 0 | 0 | 1 |
| 1 | 2 | 1 | 1 | Cumings, Mrs. John Bradley (Florence Briggs Th... | 1 | 38.0 | 1 | 0 | PC 17599 | 71.2833 | 1 | 0 | 0 |
| 2 | 3 | 1 | 3 | Heikkinen, Miss. Laina | 1 | 26.0 | 0 | 0 | STON/O2. 3101282 | 7.9250 | 0 | 0 | 1 |
| 3 | 4 | 1 | 1 | Futrelle, Mrs. Jacques Heath | 1 | 35.0 | 1 | 0 | 113803 | 53.1000 | 0 | 0 | 1 |

## 5. Normalization

```
Minimum values:
Age       0.0
Fare      0.0
SibSp     0.0
Parch     0.0
dtype: float64

Maximum values:
Age       1.0
Fare      1.0
SibSp     1.0
Parch     1.0
dtype: float64
```

# 6. Standardization

```
Mean values:
Age         2.272780e-16
Fare        3.987333e-18
SibSp       4.386066e-17
Parch       5.382900e-17
dtype: float64

Standard deviation:
Age         1.0
Fare        1.0
SibSp       1.0
Parch       1.0
dtype: float64
```

# 7. Processed Dataset Preview

| | PassengerId | Survived | Pclass | Name | Sex | Age | SibSp | Parch | Ticket | Fare | Embarked_C | Embarked_Q | Embarked_S |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 0 | 3 | Braund, Mr. Owen Harris | 0 | 22.0 | 1 | 0 | A/5 21171 | 7.2500 | 0 | 0 | 1 |
| 1 | 2 | 1 | 1 | Cumings, Mrs. John Bradley (Florence Briggs Th... | 1 | 38.0 | 1 | 0 | PC 17599 | 71.2833 | 1 | 0 | 0 |
| 2 | 3 | 1 | 3 | Heikkinen, Miss. Laina | 1 | 26.0 | 0 | 0 | STON/O2. 3101282 | 7.9250 | 0 | 0 | 1 |
| 3 | 4 | 1 | 1 | Futrelle, Mrs. Jacques Heath | 1 | 35.0 | 1 | 0 | 113803 | 53.1000 | 0 | 0 | 1 |